Project Report: E-Commerce Purchase Prediction System

Contents

Project Report: Loan Default Prediction System 1

1. Data Understanding Report 4

1.1 Problem Statement 4

Business Context 4

1.2 Objective 4

1.3 Target Variable 4

1.4 Input Variables 4

1.5 Initial Data Checks 5

Observations 5

1.6 Data Cleaning Summary 5

Unique Columns Removed 5

Null Value Handling 5

Duplicate Handling 6

Error Value Handling 6

Column Segregation 6

1.7 Outlier Detection & Treatment 6

Technique Used: IQR Method 6

Action 6

1.8 Exploratory Analysis Summary 6

Univariate Analysis 6

Bivariate Analysis 7

Key Observations 7

2. Feature Engineering Documentation 7

2.1 Categorical Encoding 7

Techniques Used: 7

2.2 Feature Transformation 7

2.3 Feature Selection 8

Removed: 8

2.4 Final Feature Set 8

3. Model Development and Evaluation Report 8

3.1 Dataset Splitting 8
3.2 Models Implemented 8
1. Logistic Regression 8
2. Decision Tree 8
3. Random Forest 9
3.3 Model Training 9
3.4 Model Evaluation 10
Metrics Used (Classification) 10
3.5 Results Summary 10
3.6 Model Selection 10
Reason 10
4. Insights and Recommendations Report 10
4.1 Key Insights 10
4.2 Feature Importance (Example) 11
4.3 Business Recommendations 11
1. Strengthen Credit Evaluation 11
2. Risk-Based Loan Approval 11
3. Income-Based Filtering 11
4. Customized Loan Products 11
4.4 Model Usage 12
4.5 Future Enhancements 12
5. Conclusion 12

1. Data Understanding Report

1.1 Problem Statement

The objective of this project to analyse historical customer data and applying classification algorithms, the model predicts whether a user is likely to make a purchase. The output is categorized into two classes: **1 (Likely to Purchase)** and **0 (Not Likely to Purchase)**, enabling businesses to take targeted actions based on predicted behavior.

.

Business Context

The e-commerce industry faces the challenge of low conversion rates despite high website traffic.Many users browse products but do not complete purchases, leading to missed revenue opportunities. Businesses invest heavily in marketing, but without proper targeting, it increases costs and reduces efficiency. A predictive system can help identify potential buyers based on their behavior. This enables companies to optimize marketing strategies, improve customer experience, and increase overall sales.

1.2 Objective

- Predict whether a customer will make a purchase (0/1)
- Identify key features influencing buying behavior
- Improve marketing and targeting strategies
- Enhance customer engagement and conversion rates

1.3 Target Variable

The target variable is '**Purchase**', which is binary: **1** for **purchase** and **0** for **no purchase**.

1.4 Input Variables

Typical features include:

- Age
- Gender
- Location

-

- Device_Type
- Time_Spent
- Pages_Visited
- Product_Category
- Previous_Purchases
- Avg_Order_Value
- Discount

1.5 Initial Data Checks

Performed:

- `df.shape` → Dataset dimensions
- `df.info()` → Data types and null values
- `df.describe()` → Statistical summary
- `df.head()` / `df.tail()` → Sample inspection

Observations

- Presence of missing values
- Mix of categorical and numerical variables
- Target variable imbalance possible

1.6 Data Cleaning Summary

Unique Columns Removed

No unique columns present in the dataset.

Null Value Handling

Column Type	Strategy
Numerical (float)	Mean Imputation
Numerical (int)	Median Imputation
Categorical	Mode Imputation

Duplicate Handling

- Checked using `df.duplicated().sum()`
- Removed using `df.drop_duplicates()`

Error Value Handling

- Replaced values like `"?"`, `"unknown"` with `NaN`
- Treated as missing values

Column Segregation

- Categorical columns
- Numerical columns

1.7 Outlier Detection & Treatment

Technique Used: IQR Method

- Lower Bound = Q1 - 1.5 * IQR
- Upper Bound = Q3 + 1.5 * IQR

Action

- Removed extreme outliers beyond bounds

1.8 Exploratory Analysis Summary

Univariate Analysis

- Distribution of age,times_spent,pages_visited,Previous_purchases,avg_order_value,discount,
- Frequency of categorical variables (countplot)

-

Bivariate Analysis

Features like Time_Spent, Pages_Visited, and Previous_Purchases show a strong positive relationship with purchase.

Key Observations

- Customers who spend more time on the website and visit more pages are more likely to make a purchase.
- Users with previous purchase history show higher chances of buying again (loyal customers).
- Discounts positively influence customer decisions, increasing purchase probability.
- Certain product categories and device types (especially mobile users) contribute more to purchases.

---

2. Feature Engineering Documentation

2.1 Categorical Encoding

Techniques Used:

Type	Method
Binary categorical	Label Encoding
Ordinal	Ordinal Encoding
Nominal	One-Hot Encoding

2.2 Feature Transformation

- Log transformation applied (if skewed)
- Scaling applied where required (for linear models)

2.3 Feature Selection

Removed:

- Non-informative features (Loan_ID)
- Highly correlated features (if multicollinearity detected)

2.4 Final Feature Set

- Cleaned and transformed dataset used for modeling
- Input matrix: X
- Target variable: y

---

3. Model Development and Evaluation Report

3.1 Dataset Splitting

- Train-Test Split:
  - 80% training
  - 20% testing

Outputs:

- X_train, X_test
- y_train, y_test

3.2 Models Implemented

1. Logistic Regression

- Baseline model
- Interpretable

2. Decision Tree

- Handles non-linearity

-

- Easy to visualize

3. Random Forest

- Ensemble model
- Reduces overfitting

3.3 Model Training

Models trained using:

- `model.fit(X_train, y_train)`

3.4 Model Evaluation Metrics Used (Classification)

- Confusion Matrix
- Accuracy
- Precision
- Recall
- Misclassification Rate

3.5 Results Summary

Model	Accuracy	Precision	Recall
Logistic Regression	78%	75%	72%
Decision Tree	80%	77%	76%
Random Forest	85%	83%	81%

3.6 Model Selection

- **Selected Model: Random Forest**

Reason

- Highest accuracy
- Better generalization
- Handles feature interactions well

---

4. Insights and Recommendations Report

4.1 Key Insights

- Customer engagement features like **Time_Spent** and **Pages_Visited** are strong predictors purchase behavior.
- Customers with **previous purchase history** have a higher likelihood of making repeat purchases.
- **Discounts and offers** significantly increase the probability of purchase.

-

- Certain **product categories and device types (especially mobile users)** show higher conversion rates.

4.2 Feature Importance (Example)

Top contributing features:

- Time_Spent
- Pages_Visited
- Previous_Purchases
- Discount

4.3 Business Recommendations

1. Target High-Intent Users:

• Focus marketing efforts on customers who spend more time and visit more pages, as they have a higher probability of purchase

---

2. Personalized Offers:

• Provide customized discounts and product recommendations based on user behavior and past purchases to increase conversions.

---

3. Customer Retention Strategies:

• Encourage repeat purchases by offering loyalty programs and exclusive deals to existing customers.

---

4. Mobile Optimization:

• Since many users access through mobile devices, improve mobile app/website experience to boost engagement and sales.

4.4 Model Usage

Input: Customer details

- Output: Purchase Prediction (0/1)

4.5 Future Enhancements

- Real-time user behavior integration
- Use of advanced models (XGBoost, Neural Networks)
- Deployment in live e-commerce system
- Customer segmentation for personalization

---

5. Conclusion

This project successfully developed a machine learning model to predict customer purchase behavior in an e-commerce environment. By analyzing user demographics, browsing patterns, and transaction history, the model was able to identify key factors influencing purchase decisions. Among the implemented models, Random Forest achieved the best performance, providing reliable and accurate predictions. The insights gained from this analysis can help businesses improve targeting strategies, enhance customer experience, and increase conversion rates. Overall, the project demonstrates the effectiveness of data-driven approaches in solving real-world business problems.

-